

SAHAN GUNAWARDENA

Software Engineer

Panadura, Sri Lanka | +94 76 194 3480 | sahan.s.gunawardena@gmail.com
linkedin.com/in/sahan-gunawardena | github.com/Sahan-Gunawardena | [Portfolio](#)

PROFESSIONAL SUMMARY

Results-driven Software Engineer with 1+ year of industry experience building and maintaining production-level web applications using PHP (CodeIgniter), MySQL, and JavaScript. Skilled in full-stack development across the MERN stack, Spring Boot, and FastAPI, with additional hands-on experience in Flutter mobile development and Python-based Machine Learning. Strong collaborator with a consistent record of delivering features on time, improving code quality, and contributing to real client projects from requirements gathering through deployment.

TECHNICAL SKILLS

Programming Languages: Java, Python, C/C++, PHP, JavaScript

Frontend Development: HTML5, CSS3, React.js, jQuery

Backend Development: Node.js, Express.js, PHP (CodeIgniter), Spring Boot, FastAPI

Databases: MySQL, MS SQL Server, MongoDB, Firebase

Mobile Development: Flutter, Kotlin (Android), React Native

Cloud & DevOps: AWS, Azure, Git, GitHub, Linux (Ubuntu), Postman, Swagger

Machine Learning: Python ML Libraries (scikit-learn, TensorFlow), Image Processing, FastAPI ML Integration

PROFESSIONAL EXPERIENCE

Freelance Software Engineer – Research & Web Projects

Mar 2025 – Present

Self-Employed

- Develop and deliver end-to-end web applications using the MERN stack (MongoDB, Express.js, React.js, Node.js) for research and commercial clients.
- Design and implement RESTful backend APIs using FastAPI to integrate Machine Learning models with web and mobile frontends.
- Perform data preprocessing, feature engineering, and supervised model training using Python-based ML libraries.
- Collaborate with clients and research teams to define technical requirements, system architecture, and project deliverables.
- Maintain clean, well-documented codebases and enforce version control best practices using Git/GitHub.

Trainee Software Engineer

Dec 2024 – Feb 2025

Earrow (Pvt) Ltd

- Developed and maintained production-level features using PHP (CodeIgniter) and MySQL, delivering modules within sprint deadlines.
- Improved database query efficiency through SQL refactoring, reducing average response times and increasing application reliability.
- Configured and supported Linux (Ubuntu) development environments, including system maintenance and deployment tasks.
- Collaborated with a cross-functional team during agile development cycles to ensure timely feature delivery.
- Applied Git version control best practices and contributed to internal knowledge-sharing sessions.

Intern Software Engineer

Apr 2024 – Oct 2024

Earrow (Pvt) Ltd

- Assisted in developing web application modules for a Higher Education Management System using PHP, CodeIgniter, MySQL, jQuery, HTML, and CSS.
- Participated in client requirement gathering sessions and supported documentation of technical specifications.
- Identified and resolved UI bugs and supported integration of frontend components with backend APIs under senior developer guidance.
- Gained practical experience in MVC architecture, relational database design, and agile software development practices.

EDUCATION

BSc (Hons) in Information Technology

2021 – 2025

Sri Lanka Institute of Information Technology (SLIIT)

G.C.E. Advanced Level

2020

Mahanama College, Colombo 03

Combined Mathematics: S | Physics: S | Chemistry: C

G.C.E. Ordinary Level

2016

Mahanama College, Colombo 03

Results: 8 A's and 1 B

PROJECTS

Banana Leaf Disease Detection Mobile Application | *Python, Machine Learning, FastAPI, Flutter, Image Processing*

Built a full-stack mobile app to detect banana leaf diseases in real time. Developed a FastAPI backend that processes uploaded images and returns ML-generated disease predictions; created the Flutter frontend to capture images via camera or gallery and display results instantly.

Biomechanical Analysis for Weightlifting Performance Enhancement | *Machine Learning, 3D Modelling, Mobile Development*

Developed a mobile application that performs real-time analysis of weightlifting training videos. Implemented 3D visualisation models of correct technique and delivered personalised performance recommendations and corrective feedback to users.

Online Garbage Collecting System | *MERN Stack - React.js, Node.js, Express.js, MongoDB*

Built the Leave Management module within a full garbage-collection platform, enabling staff to submit leave requests, track approval status, and view leave history.

Online Shopping App for Visually Challenged Users | *Kotlin, Android Studio, Google Firebase*

Developed dynamic theme-adaptation features that automatically adjust the app interface - font size, contrast, and layout-based on the user's individual visual impairment profile.

Social Media Web Application for Gym Community | *Spring Boot, Java*

Contributed to a social media platform for gym members to post updates, share training tips, and engage with the community. Implemented backend REST endpoints and business logic using Spring Boot.

UX/UI Website Redesign | *Figma*

Conducted UX audit of an existing website, identified usability and accessibility issues, and prototyped a redesigned interface in Figma with improved information architecture and visual hierarchy.